

remember learning in geometry class that the sine is defined in terms of triangle lengths: the length of the angle's opposite side divided by the length of the hypotenuse. But if that is the definition of a sine, how can a calculator have a sin button that performs this calculation instantaneously? Does the calculator draw a right triangle in its innards, get out a ruler and measure the lengths of the sides, and then divide them? We might ask a similar question for square roots: the square root is the inverse of a square, and there's no straightforward, closed-form arithmetic formula for it that a calculator could use. I imagine that you can already guess the answer: there is an algorithm for quick calculations of square roots.

The Babylonian Algorithm

Suppose that we need to find the square root of a number x . As with any math problem, we can try a guess-and-check strategy. Let's say that our best guess for the square root of x is some number y . We can calculate y^2 , and if it's equal to x , we're done (having achieved a rare completion of the one-step "lucky guess algorithm").

If our guess y is not exactly the square root of x , then we'll want to guess again, and we'll want our next guess to take us closer to the true value of the square root of x . The Babylonian algorithm provides a way to systematically improve our guesses until we converge on the right answer. It's a simple algorithm and requires only division and averaging:

1. Make a guess, y , for the value of the square root of x .
2. Calculate $z = x/y$.
3. Find the average of z and y . This average is your new value of y , or your new guess for the value of the square root of x .
4. Repeat steps 2 and 3 until $y^2 - x$ is sufficiently small.

We described the Babylonian algorithm in four steps. A pure mathematician, by contrast, might express the entire thing in one equation: